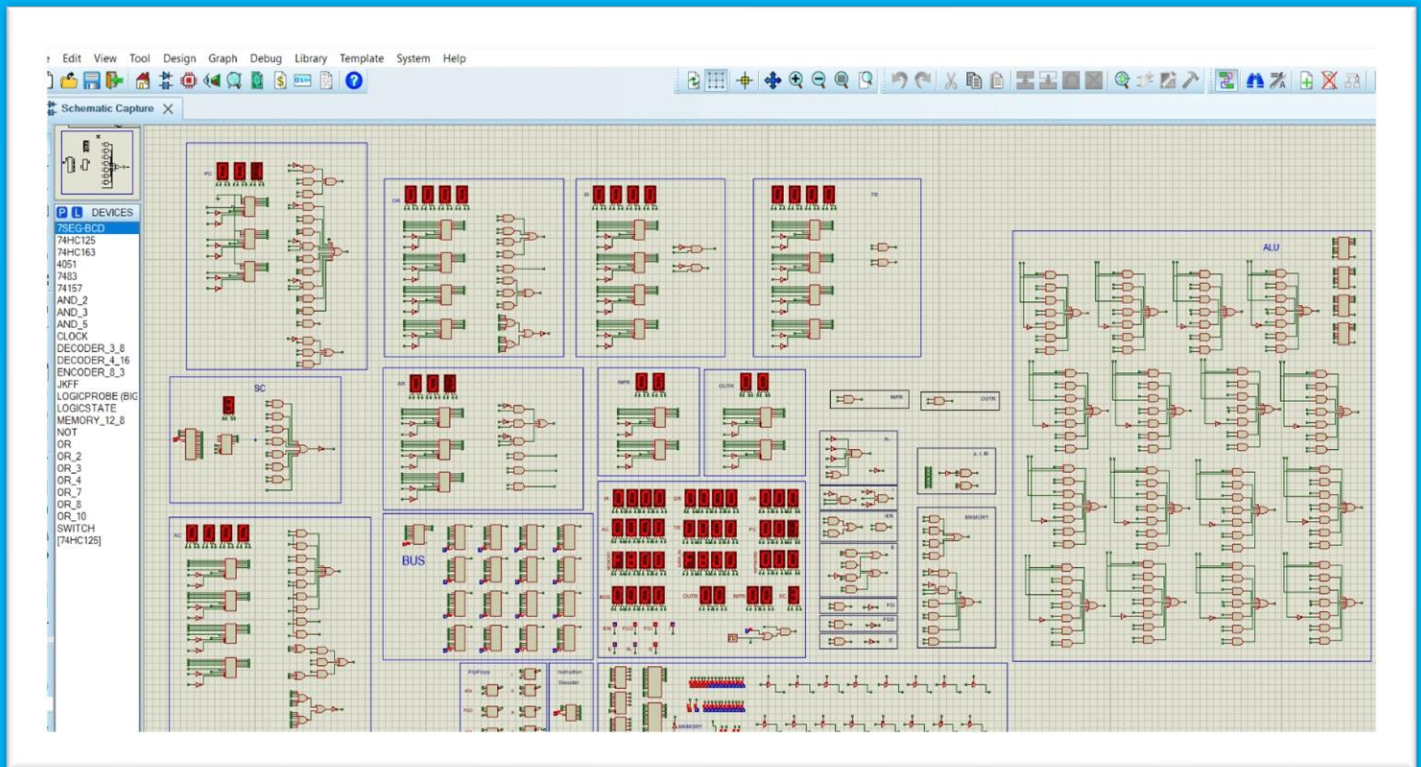


Mano_Basic_Computer _Design

Innovators microprocessors team

Picture of Mano_Basic_Computer_design



Looking at the schematic, we can discern several key functional blocks that correspond to the essential components of the Mano Basic Computer:

- **Registers:** Multiple rectangular blocks with numerous input and output lines are clearly visible. These represent the various registers within the CPU, such as the Program Counter (PC), Address Register (AR), Data Register (DR), Accumulator (AC), Instruction Register (IR), Temporary Register (TR),¹ and potentially input (INPR) and output (OUTR) registers. The individual lines connected to these blocks signify the parallel bit lines used for storing and transferring binary data.
- **Memory:** A larger, more intricate block likely represents the main memory (RAM) of the computer. It will have address inputs to select specific memory locations and data input/output lines for reading and writing information.

- **Arithmetic Logic Unit (ALU):** Another distinct block is dedicated to the ALU. This crucial component performs arithmetic (addition, subtraction, etc.) and logical (AND, OR, NOT, XOR) operations on the data as directed by the control unit.
- **Control Unit Logic:** The most complex and distributed part of the design appears to be the control unit. This is likely implemented using various smaller interconnected blocks representing combinational logic circuits (like AND, OR, NOT gates), decoders, multiplexers, and sequential logic circuits (like flip-flops). The control unit's function is to generate the precise sequence of control signals needed to orchestrate the operations of all other components based on the instruction being executed.
- **Bus System:** While not always explicitly drawn as a single thick line, the numerous interconnected wires running throughout the schematic indicate the presence of a bus system. This set of parallel conductors allows for the efficient transfer of data between different registers, the memory, and the ALU.
- **Input/Output (I/O) Interface:** Certain sections of the schematic likely handle the interaction with the outside world. This could include input mechanisms (simulated switches or input devices) and output mechanisms (like LEDs or simulated displays) for observing the results of the computer's operations.
- **Clock and Timing Circuitry:** Although not always a prominent block, there will be components generating the necessary clock pulses. These timing signals are essential for synchronizing the sequential operations within the digital computer.
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- - **The main function of this project is to simulate the fundamental instruction cycle of the Mano Basic Computer:**
1. **Fetch:** The control unit retrieves an instruction from memory. The address of the instruction is held in the Program Counter (PC). This address is sent to the memory via the Address Register (AR). The instruction is then read from memory into the Data Register (DR) and subsequently transferred to the Instruction Register (IR). The PC is then incremented to point to the next instruction in sequence.
 2. **Decode:** The control unit analyzes the opcode (the operational part) of the instruction stored in the IR. This decoding process determines what operation needs to be performed.
 3. **Execute:** Based on the decoded instruction, the control unit generates a specific sequence of control signals. These signals activate the appropriate components – the ALU for calculations, registers for data storage, and memory for reading or writing – to carry out the instruction. This step can involve arithmetic operations, logical operations, data transfers between registers and memory, or input/output operations.

This cycle then repeats, with the computer fetching and executing the next instruction in memory.

In essence, this Proteus project provides a visual and functional model of a basic computer architecture. By simulating the interaction of its core components, it allows learners to understand the fundamental principles behind how a computer processes information and executes programs at a hardware level.

Observing the signals and data flow within this simulated environment offers valuable insights into the inner workings of a central processing unit.

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